Donna L. Hoffman & Thomas P. Novak

Marketing in Hypermedia Computer-Mediated Environments: Conceptual Foundations

The authors address the role of marketing in hypermedia computer-mediated environments (CMEs). Their approach considers hypermedia CMEs to be large-scale (i.e., national or global) networked environments, of which the World Wide Web on the Internet is the first and current global implementation. They introduce marketers to this revolutionary new medium, propose a structural model of consumer navigation behavior in a CME that incorporates the notion of flow, and examine a series of research issues and marketing implications that follow from the model.

Firms communicate with their customers through various media. Traditionally, these media follow a passive one-to-many communication model, whereby a firm reaches many current and potential customers, segmented or not, through marketing efforts that allow only limited forms of feedback from the customer. For several years, a revolution has been developing that is dramatically altering this traditional view of advertising and communication media. This revolution is the Internet, the massive global network of interconnected packet-switched computer networks, which as a new marketing medium has the potential to radically change the way firms do business with their customers.

The Internet operationalizes a model of distributed computing that facilitates interactive multimedia many-to-many communication (for a complete history, see Hafner and Lyon 1996). As such, the Internet supports discussion groups (e.g., USENET news, moderated and unmoderated mailing lists), multiplayer games and communication systems (e.g., MUDs [Multiuser Dungeons], IRC [Internet Relay Chat], chat, MUSEs [Multiuser Shared Environment]), file transfer, electronic mail, and global information access and retrieval systems (e.g., archie, gopher, World Wide Web).[1] The business implications of this model, in which "the engine of democratization sitting on so many desktops is already out of control, is already creating new players in a new game" (Carroll 1994, p. 73), will be played out in as yet unknown ways for many years to come.

We focus on the marketing implications of commercializing hypermedia computer-mediated environments (CMEs), of which the World Wide Web (Berners-Lee et al. 1993) on the Internet is the first and current networked global implementation. Although we provide a formal definition of a hypermedia CME subsequently, at this point we informally define it as a distributed computer network used to access and provide hypermedia content (i.e., multimedia content connected across the network with hypertext links). Although other CMEs are relevant to marketers, including private bulletin board systems; conferencing systems such as The WELL (Rheingold 1992); and commercial on-line services such as America Online, CompuServe, Prodigy, and the Microsoft Network, we restrict our focus to marketing activities in hypermedia CMEs accessible via the Web on the Internet.

From a commercial perspective, the Web consists of locations, or sites, that firms erect on servers and consumers

Donna L. Hoffman and Thomas P. Novak are Associate Professors, Owen Graduate School of Management, Vanderbilt University. This research is based on work supported by the National Science Foundation under Grant No. SBR-9422780, Interval Research Corporation, Sun Microsystems, and the Dean's Fund of the Owen Graduate School of Management. The authors acknowledge the many helpful comments of seminar participants at the Owen School and at Interval Research Corporation, as well as those of three anonymous *JM* reviewers and the editor. The authors contributed equally to this article.

[1]For a discussion of the technical terms used in this paper, consult one of the many reference books on the Internet (e.g., December and Randall 1995). A comprehensive listing of books about the Internet can be found at http://www.switch.ch/switch/Internet-Books.txt. The http address is a Uniform Resource Locator (URL) that specifies the exact location of a document on the Internet. Note that URLs are used in the Reference list to indicate where on-line versions of documents can be identified. Although these URLs were accurate at the time this article was written, it is possible that over time, some of these URLs may have changed or no longer exist.

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visit. On the Web, consumer-oriented network navigation consists of visiting a series of *Web sites* to search for information and/or advertising about products and services or consumer content (possibly advertiser-supported) or place an order for a product.

Consumers visit a site by directly entering its Web address in the browser or clicking on a hypertext link leading to it from some other site. Once there, consumers navigate through the site using a series of point-and-click motions with a mouse or entering textual information into pop-up windows and "fill-out forms" with keyboard strokes. From there, the consumer chooses where to go next in the site. Often, the offerings are presented as a nonlinear graphical menu or map of choices to the consumer. The navigation process continues, terminated only when the consumer jumps to another off-site hypertext link within the Web or exits the Web navigation experience entirely.

The Internet is an important focus for marketers for several reasons. First, consumers and firms are conducting a substantial and rapidly increasing amount of business on the Internet.[2] Although there is some controversy surrounding the estimates of the size of the Internet, surveys performed to date by O'Reilly, FIND/SVP, Times Mirror, and Hoffman, Kalsbeek, and Novak (1996) suggest that there are at least 10 million Internet users in the United States alone. The number of computers (hosts) connected to the Internet topped 9.47 million (Network Wizards 1996) as of January 1996 and has been doubling every year since 1982. Note that a single host supports anywhere from a single user to, in some cases, thousands of users. As of March 21, 1996, Open Market's (1996) directory of "Commercial Services on the Net" listed 24,347 firms, and there were 54,800 entries in the "Companies" directory of Yahoo's (1996) guide to the Web, with the total number of Web sites doubling approximately every two months.

Second, as Malone (1995) has argued, the market prefers the decentralized, many-to-many Web for electronic commerce to the centralized, closed-access environments provided by the on-line services. Significantly, all the major on-line services now offer Web access to their subscribers and have or are expected to announce plans to allow members to self-publish their own *home pages* on the Web, as well. Additionally, virtually all the major communications conglomerates have Web sites, as they shift their strategic orientations away from so-called interactive television applications to Web-based publishing, communication, and multimedia marketing efforts.

Third, and this follows from the first and second points, the World Wide Web represents the broader context within which other hypermedia CMEs exist. Indeed, much of the foundation we develop here is relevant to on-line services, particularly as they begin to function more as full-service Internet access providers and less as closed, proprietary networks. Thus, restricted on-line services are special cases of the open-access World Wide Web. Open access results in lower entry barriers so that virtually anyone can both access and provide content to the Internet. In essence, the Web "levels the playing field."

Both the Web and proprietary on-line services are examples of developments in *electronic commerce,* which, as Rangaswamy and Wind (1994) have noted, include such developments as EDI (electronic data interchange), kiosks, electronic classified advertisements, and on-line services such as CompuServe and Minitel, the French videotex system (Cats-Baril and Jelassi 1994). With the possible exception of EDI, which is moving to the Internet because its "open architecture" system is more inclusive and offers numerous advantages over private networks, none of these mechanisms for facilitating commerce electronically has the same far-reaching scope and potential for transformation of the business function as does the World Wide Web.

Fourth, the Web provides an efficient channel for advertising, marketing, and even direct distribution of certain goods and information services. For example, Verity and Hof (1994) suggest that it may be nearly one-fourth less costly to perform direct marketing through the Internet than through conventional channels. Neece (1995) reports that SunSolve Online™ has saved Sun Microsystems over $4 million in "FAQs"[3] alone since Sun "reengineered information processes around the WWW [Web]." A recent study by IBM Corporation (1995) suggests that on-line catalogs on the Internet can save firms up to 25% in processing costs and reduce cycle time by up to 62%. Along with the suspected increases in efficiency, the anecdotal evidence mounts that marketing on the net also may be more effective than marketing through traditional media. For example, one vendor estimated that his marketing efforts on the Web resulted in "10 times as many units [sold] with 1/10 the advertising budget" (Potter 1994).

Goals and Organization of the Paper

Although there have been recent scholarly efforts detailing the impact of new information technologies on marketing

[2]The most recent figures decision makers are using for business planning and research purposes are that the total core economy for electronic commerce on the Internet will approach $45.8 billion by the year 2000 (Modahl 1995), including $14.2 billion for infrastructure, $2.8 billion for consumer content, $6.9 billion for business content, and $21.9 billion in on-line trade (made up of $6.9 billion in retail and $15 billion in EDI). Additionally, Modahl (1995) forecasts that $46.2 billion in consumer assets will be managed over the Internet by the year 2000, broken down as $29.9 billion in mutual funds and $16.3 billion in deposits. However, analysts at Hambrecht and Quist (1995, www.hambrecht.com) estimate a much larger Internet economy of $73 billion by the year 2000, comprised of $13 billion in infrastructure, $10 billion in content, and $50 billion in transactions. On the other hand, Alex, Brown and Sons (Clark 1995) forecast a much smaller Internet economy, reaching somewhere between $13.5 billion and $16 billion by 1998, composed of a $6 billion infrastructure, $5 billion to $7 billion in communication access, and $2.5 billion in content. Currently, the total Internet economy is estimated at $2 billion (Modahl 1995). Additionally, the market for the largely digital electronic convergence of content systems, publishing, and so on is already estimated at over $1 trillion (Oliver 1995). None of these forecasts include efficiency savings from moving processes and transactions on-line.

[3]A FAQ is a frequently asked question to which standardized replies are both readily available and desirable.

(e.g., Blattberg, Glazer, and Little 1994; Glazer 1991), the existing research does not discuss the impact that hypermedia CMEs such as the World Wide Web have on marketing theory and business practice. Despite the massive amount of attention given to the Internet in the popular press (see, e.g., the more than 6000 references to the Internet in *ABI Inform* through November 1995) and the belief in many business circles that the Web represents a phenomenal marketing opportunity, virtually no scholarly effort has been undertaken by marketing academics to understand hypermedia CMEs, both as media for marketing communications and as markets in and of themselves. We draw from the relevant literatures in psychology, communications, media studies, organizational behavior, and human-computer interaction, and concentrate our efforts on developing a conceptual foundation for understanding consumer behavior in hypermedia CMEs.

Therefore, our goals are to introduce marketers to this revolutionary new medium, propose a preliminary process model of consumer navigation behavior in a hypermedia CME, examine the research issues that correspond to the process model, and derive marketing implications for electronic commerce all to stimulate critical inquiry in this emerging area. To that end, the article is organized as follows: We discuss three models of communication that underlie traditional and new media and develop a media typology that reveals new insight into the distinctions among traditional media and CMEs. Then, using an expanded concept of flow (Csikszentmihalyi 1977, 1990), we introduce a preliminary process model of network navigation in CMEs. We next present a series of 15 research issues involving the flow construct and process model, while paying attention to the marketing implications that follow from the research issues. Finally, we offer conclusions about the importance of this emerging area of inquiry to both marketing scholars and practitioners interested in electronic commerce.

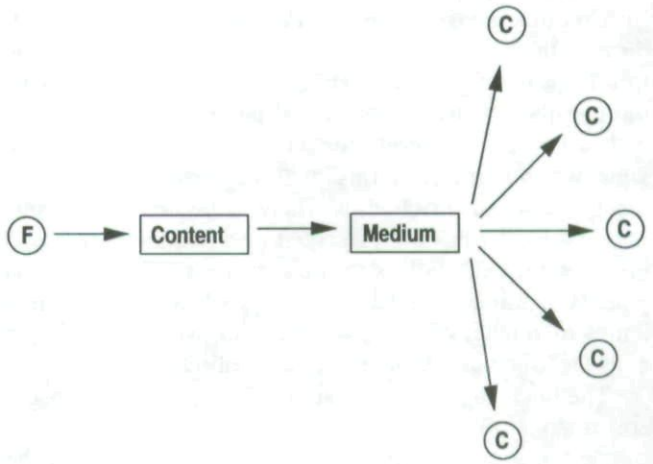
Hypermedia Computer-Mediated Environments

We begin by outlining a series of three communication models that serve to identify several unique characteristics of hypermedia CMEs such as the Web. We then describe a new media typology that positions the Web in the broader context of new and traditional media and discuss the marketing implications of the communication models and typology.

Communication Models

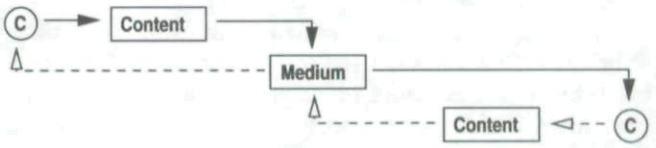
Model 1: Mass media. In Figure 1, we present a simplified model that underlies many models of mass communication (e.g., Katz and Lazarsfeld 1955; Lasswell 1948). The primary feature of Figure 1 is a one-to-many communications process, whereby the firm (F) transmits content through a medium to consumers (C). Depending on the medium (i.e., broadcast, print, and billboards), either static (i.e., text, image, and graphics) and/or dynamic (i.e., audio, full-motion video, and animation) content can be incorporated. No interaction between consumers and firms is present in this model. Virtually all contemporary models of mass media effects are based on this traditional model of the communication process (e.g., see Kotler 1994, Chapter 22).

FIGURE 1
Traditional One-To-Many Marketing Communications Model for Mass Media



Note: F = firm; C = consumer.

FIGURE 2
Model of Interpersonal and Computer-Mediated Communication



Note: C = consumer.

Model 2: Interpersonal and computer-mediated communications. In Figure 2, we present a simplified model of interpersonal communication that is based on traditional models of communication from sender to receiver. The solid and dashed lines indicate communication flows through a medium for two distinct persons. This model incorporates a feedback view of interactivity, which is consistent with Rafaeli's (1988, p. 111) definition of interaction as "an expression of the extent that in a given series of communication exchanges, any third (or later) transmission (or message) is related to the degree to which previous exchanges referred to even earlier transmissions." Although Figure 2 is shown here for one-to-one communication between two consumers, the model can be easily extended to represent many-to-many interpersonal communication (i.e., teleconference, face-to-face group meetings, or on-line chat rooms). Note that unmediated face-to-face interpersonal communication is a special case of Figure 2. From a marketing perspective, the model in Figure 2 is implicit in developments of word-of-mouth communication models.

Person-interactivity, the key feature distinguishing Figure 2 from Figure 1, is defined as interactivity between peo-

ple that occurs through a medium or is unmediated, as in the case of face-to-face communication. In this view of interactivity, media are "important only as a conduit, as a means of connecting sender and receiver, and are only interesting to the extent that they contribute to or otherwise interfere with the transmission of messages from sender to receiver" (Steuer 1992, p. 77–78).

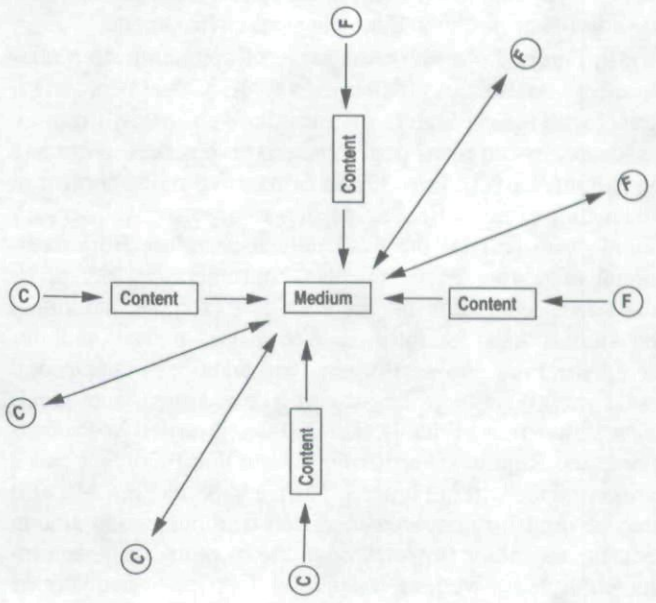
Hypermedia CMEs. Nearly 50 years ago, Busch (1945, p. 106) proposed a hypertext-like system called *Memex*, which would consist of "a device in which an individual stores all his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility." Nelson (1967) discussed hypertext in terms of a network of paths and associations, with an emphasis on approximating the way the human brain connects information. Bornman and von Solms (1993, p. 260) provide a current definition: "*Hypertext* suggests the concept of non-sequential writing of information that allows the user to connect information together by means of different paths or links." *Multimedia* uses a computer to integrate and provide interactive access to both static (i.e., text, image, and graphics) and dynamic (i.e., audio, full-motion video, and animation) content, whereas hypermedia combines the node-and-link access of hypertext with multimedia content. Gygi (1990) and Smith and Wilson (1993) provide more extensive discussion of hypertext and hypermedia.

We define a *hypermedia CME* as a dynamic distributed network, potentially global in scope, together with associated hardware and software for accessing the network, which enables consumers and firms to (1) provide and interactively access hypermedia content (i.e., "machine interactivity") and (2) communicate through the medium (i.e., "person interactivity"). We further define *network navigation* as the process of self-directed movement through a hypermedia CME. This nonlinear search and retrieval process provides essentially unlimited freedom of choice and greater control for the consumer and may be contrasted with the restrictive navigation options available in traditional media such as television or print. Furthermore, network navigation permits much greater freedom of choice than the centrally controlled interactive multimedia systems, such as video-on-demand and home-shopping applications of so-called Interactive Television; the text-based French Minitel system (Cats-Baril and Jelassi 1994); the menu-based information-acceleration approach of Hauser, Urban, and Weinberg (1993); or the experimental systems for monitoring information processing, such as Mouselab (e.g., Payne, Bettman, and Johnson 1993).

Model 3: A new model for hypermedia CMEs. In Figure 3, we present a many-to-many communication model for hypermedia CMEs. The content in Figure 3 is hypermedia, and the medium is a distributed computer network. Figure 3 differs from Figure 2 in that interactivity can also be *with* the medium (i.e., machine interactivity) in addition to *through* the medium (i.e., person interactivity).

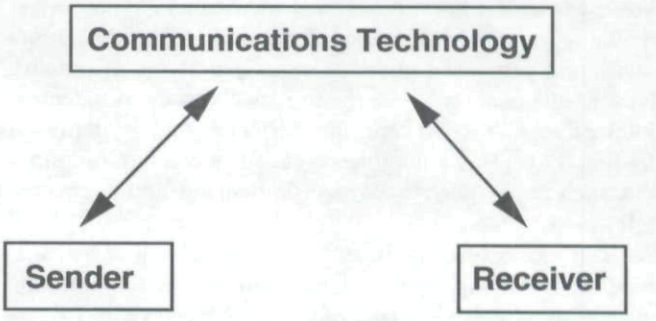
Figure 3 is based on a communication model outlined by Steuer (1992) and shown in Figure 4. The mediated model

FIGURE 3
A Model of Marketing Communications in a Hypermedia CME



Note: F = firm; C = consumer.

FIGURE 4
Mediated Communication



Note: Adapted from Steuer (1992).

represented in Figure 4 suggests that the primary relationship is not between the sender and the receiver, but rather with the mediated environment with which they interact. According to this view, information or content is not merely transmitted from a sender to a receiver; instead, "mediated environments are created and then experienced" (Steuer 1992, p. 78). In Steuer's model of mediated communication, *machine interactivity* is "the extent to which users can participate in modifying the form and content of a mediated environment in real time" (p. 84). Steuer calls his model a *telepresence view* of mediated communication, where *presence* is defined as "the natural perception of an environment" and *telepresence* is defined as "the mediated perception of an environment" (p. 76). Following Steuer, when in-

teracting with a computer-mediated environment, the consumer perceives two environments: (1) the physical environment in which he or she is present and (2) the environment defined by the hypermedia CME. The strength of the experience of telepresence is a function of the extent to which a person feels present in the hypermedia CME, rather than in his or her immediate physical environment.

In Figure 3, we show the range of communication relationships possible in a hypermedia CME. Consumers can interact with the medium (e.g., "surf the Web" using browsing software) as can firms (e.g., business-to-business marketing in CommerceNet). In addition, firms can provide content to the medium (e.g., a firm establishes a corporate Web server). Finally—in perhaps the most radical departure from traditional marketing environments—consumers can put product-related content in the medium. For example, individual consumers have established Web pages for automobiles (e.g., the Ford Probe, Porsche, car audio equipment, and solar cars), toys (e.g., Legos and Barbie Dolls), and television shows (e.g., Friends, The X-Files, Married With Children, and Rugrats). Furthermore, note that Figures 1 and 2 are contained within Figure 3. Thus, a hypermedia CME also can be used for computer-mediated communication among consumers and/or firms (*through* the medium, or person interactivity), as well as potentially for one-to-many mass communication, though applications of the latter have met with considerable consumer resistance (e.g., Godwin 1994).

Media Characteristics

Studying media characteristics provides a structured context for comparing different media types, including traditional media and new media (Valacich et al. 1993; Williams, Strover, and Grant 1994). Traditional media include both mass media (e.g., television, radio, newspaper, magazines, direct mail), and personal communications (e.g., word of mouth). New media encompass interactive media, such as videotex, interactive CD-ROM, on-line services, and hypermedia CMEs, as well as emerging so-called interactive multimedia, such as pay-per-view, video-on-demand, and interactive television.

The media typologies referenced in Table 1 reveal that media differ along many different dimensions, for example, channel characteristics (Reardon and Rogers 1988; Rogers 1986), social presence (Rice 1992, 1993), and uses and gratifications (Perse and Courtright 1993). However, hypermedia CMEs were not in existence at the time these typologies were proposed. Although the typologies cited include the computer as a communications medium, it is defined narrowly in terms of email, bulletin boards, and computer conferencing.

Furthermore, many of the characteristics listed in Table 1 are either subjective in nature or require the application of valid measurement procedures. Therefore, we propose a new media typology based on seven objective characteristics. Although objective characteristics do not allow the media to be compared with respect to such psychological dimensions as communication needs, social presence, or control, they do permit a relatively error-free classification. Thus, in Table 2 re, we characterize 35 traditional and new media with respect to seven objective characteristics. We have already discussed person-interactivity and machine-interactivity and the distinction between one-to-many and one-to-few (Figure 1), one-to-one and few-to-few (Figure 2), and many-to-many (Figure 3) communication models. Content simply identifies whether static or dynamic information can be delivered by the medium. For unmediated interpersonal communication, experiential content includes stimuli affecting additional sensory modalities, such as tactile, proprioceptive, or olfactory senses.

The remaining three characteristics may be defined briefly as follows: The *number of linked sources available* specifies how many sources of content are readily accessible or available to the user at any given usage opportunity. *Media feedback symmetry* refers to whether different parties in the communication process employ differing media bandwidths for sending information. For example, in an Interactive CD, feedback is asymmetric, because the Interactive CD sends high bandwidth information, but the consumer sends low bandwidth information. From the consumer's perspective, this facilitates interactivity because a few simple cursor, mouse, or joystick movements produce dramatic modifications in the environment. When there is symmetric media feedback, all sources in the communication process employ the same media bandwidth for sending information—for example, telephone, mail, and face-to-face communication. *Temporal synchronicity* is a property of interactive media only, does not apply to mass media, and means interaction occurs in real time.

In Figure 5, we present a perceptual map, produced by plotting object scores for 35 media types resulting from a nonlinear principal components analysis (Gifi 1990) of the data from Table 2. The nonlinear principal components analysis is equivalent to a multiple correspondence analysis (Hoffman and de Leeuw 1992), with ordinal restrictions imposed on category quantification of variables assumed to have a known underlying ordering of categories.[4] To simplify presentation, only the object scores for the rows of Table 2, and not the category quantifications for the columns of Table 2, are plotted.

Following orthogonal rotation of axes, Figure 5 has a clear interpretation. The horizontal axis differentiates impersonal from personal communication media, and the vertical axis differentiates dynamic from static media. Reardon and Rogers (1988, p. 297) argue that "new communication technologies are interactive in nature, and thus cannot be easily categorized as either interpersonal or mass media channels." Indeed, Figure 5 shows that traditional mass media channels occupy positions at the upper left (broadcast media) and lower left (print media), whereas interpersonal

[4]In this case, ordinal restrictions were imposed on the number of linked sources (one, few, many) and the three-category variable specifying presence of text content. The communication model also was treated as two separate ordinal variables—the number of senders (one, few, many) and the number of receivers (one, few, many). Because all other variables were either binary (person-interactivity, machine-interactivity, audio, video, image, experiential, and media feedback symmetry) or nominal (temporal synchronicity), no other ordinal restrictions were imposed.

TABLE 1
Review of Media Comparisons

Source	Commuication Media Compared	Characteristics of Communication Media
Dennis and Valacich (1994)	Face-to-face, phone, memo, voice mail, video conference, email, electronic phone, and group support systems	Feedback, symbol variety, concurrency, persistence, ability to rehearse
Perse and Courtright (1993)	Television, video cassette recorder, movies, conversation, phone, computer, newspapers, magazines, books, and radio	11 communication needs (relaxation, entertainment, forget work, friendship, learning, pass time, excitement, feel less lonely,satisfy habits, acknolwedge feelings, get someone to do something for me) and social presence
Reardon and Rogers (1988)	Interpersonal communication, interactive media, and mass media	Message flow, source knowledge of the audience, segmentation, interactivity, feedback, asychronicity, emotional versus task-related content, control, privacy
Steuer (1992)	44 new and old media including newspapers, fax, interactive television, three-dimensional films, sensorama, and the Star Trek Holodec	Subjective classification according to vividness and interactivity
Stewart and Ward (1994)	Television, radio, magazines, and newspapers	27 characteristics for "gross media comparisons"
Rice (1992)	Email, voice mail, videoconferencing, and on-line databases	Social presence, information richness
Rice (1993)	Face-to-face, email, meetings, phone, desktop video, text, and voice mail	Social presence, appropriateness for ten communication activities (exchange information, negotiate, get to know someone, ask questions, stay in touch, exchange time-sensitive information, generate ideas, resolve disagreements, make decisions, confidentiality)
Valacich and colleagues (1993)	Distributed verbal, face-to-face verbal, distributed electronic, and face-to-face electronic	Communication concurrency Kinds of information, mode of communication
van Dijk (1993)	Two-way cable, videotex, data networks, email, videophone, and interactive video	

media occupy the upper-right position and mail and fax occupy the lower-right position. New media occupy largely intermediate positions, in agreement with Reardon and Rogers (1988), who view new interactive media as combining properties of mass (impersonal) and face-to-face (personal) communication channels.

Figure 5 identifies two points for the Web, one assuming text, image, and audio content (Web alone) and one assuming text, image, audio, and video content (Web plus video). Web plus video content occupies a central position in Figure 5; that is, it is the most "typical" of all communications, sharing characteristics with a wide variety of other media types. The many-to-many communication model underlying the Web positions the Web closer to the personal side of the horizontal axis than it does traditional mass media.

The central position of the Web in Figure 5 corresponds to an important strategic interpretation of the Web as a marketing medium; namely, the Web combines elements from a variety of traditional media, yet is more than the sum of the parts. For example, broadcast media in the top left quadrant of Figure 5 provide relatively short-term exposure with low information content, whereas the print media in the bottom left quadrant provide relatively long-term exposure with high information content. Advertising strategy on the Web must account for both short-term (decision of which link to select next) and long-term (reading detailed information provided at a commercial site) exposure.

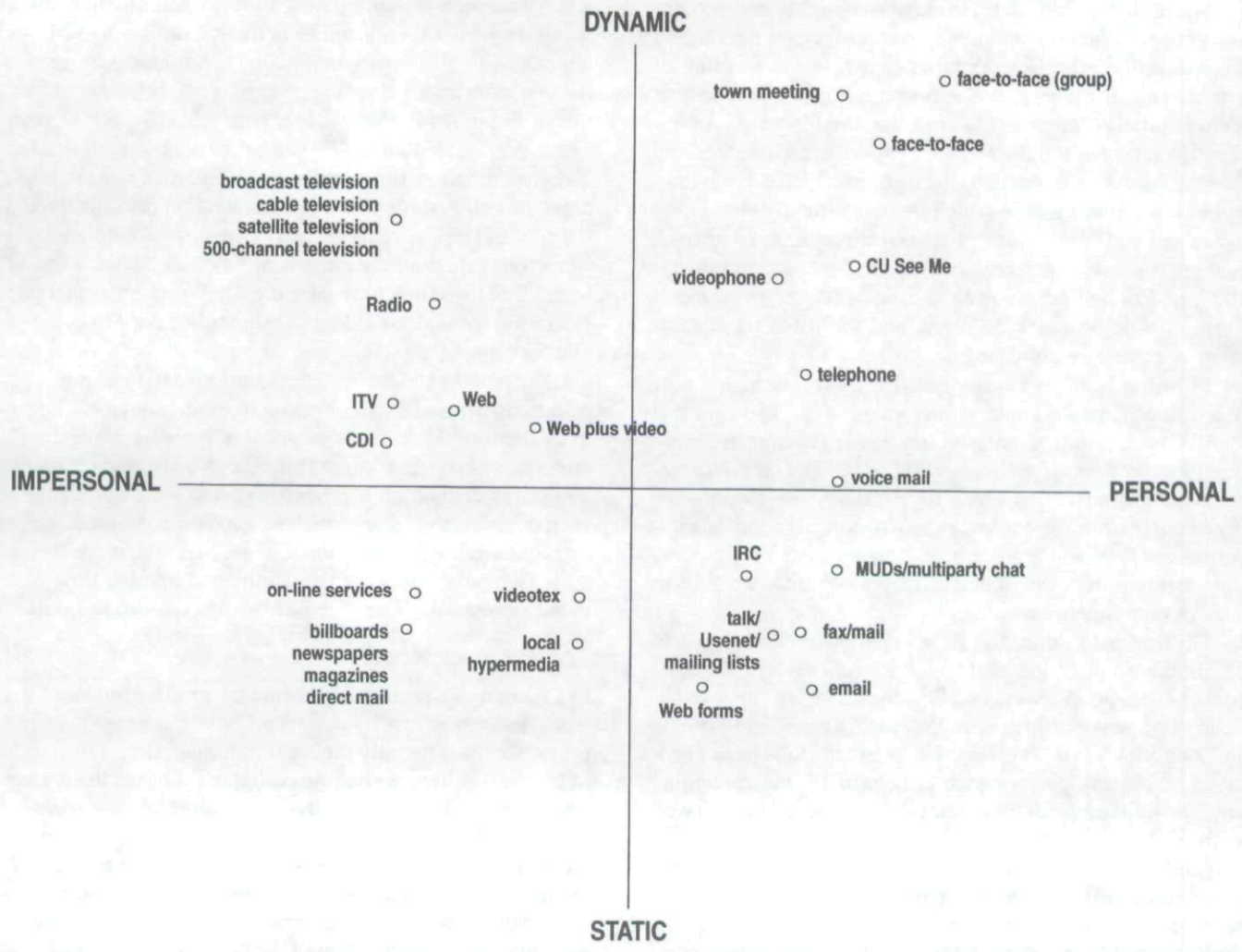
We next discuss a process model of network navigation in hypermedia CMEs such as the Web. There are several unique features of CMEs as media and marketing environments that motivate the process model. As virtual hypermedia environments incorporating interactivity with both people and computers, CMEs are not simulations of real-world environments, but alternatives to them. Within the virtual environment, both experiential (e.g., net-surfing) and goal-directed (e.g., on-line shopping) activities compete for consumers' attention. Consumer skill in functioning in the virtual environment, as well as challenges posed by the envi-

TABLE 2
Objective Characteristics of Media

	Person-interactivity	Machine-interactivity	Number of linked sources	Communication model	Content[a]	Media feedback symmetry	Temporal synchronicity
Mass Media							
Billboards	no	no	one	one-to-many	T, I	yes	n/a
Newspapers	no	no	one	one-to-many	T, I	yes	n/a
Magazines	no	no	one	one-to-many	T, I	yes	n/a
Direct mail	no	no	one	one-to-many	T, I	yes	n/a
Radio	no	no	few	one-to-many	A	no	n/a
Broadcast television	no	no	few	one-to-many	A, V, (T)	no	n/a
Cable television	no	no	few	one-to-many	A, V, (T)	no	n/a
Satellite television	no	no	many	one-to-many	A, V, (T)	no	n/a
500 channel cable television	no	no	many	one-to-many	A, V, (T)	no	n/a
Interactive Media							
Local hypertext	no	yes	one	one-to-many	T	yes	yes
Local hypermedia	no	yes	one	one-to-many	T, I, A, V	no	yes
Dial-up bulletin board service (information only)	no	yes	one	one-to-many	T	yes	yes
CD-Interactive	no	yes	one	one-to-many	T, I, A, V	no	yes
Videotex	no	yes	few	one-to-many	T	yes	yes
Pre-Web On-line Services	no	yes	few	one-to-many	T, I	no	yes
Interactive television	no	yes	few	one-to-many	T, I, A, V	no	yes
World Wide Web	no	yes	many	many-to-many	T, I, A, V	no	yes
Interpersonal Communication							
Mail	yes	no	one	one-to-one	T	yes	no
Fax	yes	no	one	one-to-one	T	yes	no
Telephone	yes	no	one	one-to-one	A	yes	yes
Videophone	yes	no	one	one-to-one	A, V	yes	yes
Face-to-face	yes	no	one	one-to-one	A, V, E	yes	yes
Face-to-face (group)	yes	no	few	few-to-few	A, V, E	yes	yes
Town meeting	yes	no	many	many-to-many	A, V, E	yes	yes
Computer-Mediated Communication							
Email	yes	yes	one	one-to-one	T	yes	no
Voice mail	yes	yes	one	one-to-one	A	yes	no
Talk program	yes	yes	one	one-to-one	T	yes	yes
Email (carbon copy: list)	yes	yes	one	one-to-few	T	yes	no
Multiparty chat	yes	yes	few	few-to-few	T	yes	yes
MUDs	yes	yes	few	few-to-few	T	yes	yes
See you see me	yes	yes	few	few-to-few	A, V	yes	yes
Mailing lists	yes	yes	many	many-to-many	T	yes	no
Usenet newsgroups	yes	yes	many	many-to-many	T	yes	no
Web (forms/annotation)	yes	yes	many	many-to-many	T, I	yes	no
Internet Relay Chat	yes	yes	many	many-to-many	T	yes	yes

[a]T = text; I = image; A = audio; V = video; E = experiential. (T) = there is a minor amount of text content.

FIGURE 5
Media Typology Based on Objective Characteristics



ronment, introduce a competency issue that does not exist in such a fundamental manner in the physical world. This issue of whether consumers' skills are competent to meet the challenges of the virtual environment plus the telepresence created by immersion in an interactive hypermedia environment bear directly on the construct of flow, which underlies the process model.

A Process Model of Network Navigation in Hypermedia CMEs

The Flow Construct

Although consumer researchers have explored the role of play in the consumption experience (e.g., Holbrook et al. 1984), we believe the concept of flow in a hypermedia CME holds wider applicability and underlies many crucial components of the consumer's interaction with the firm and its offerings. Simply stated, *flow* is the "process of optimal *experience*" (Csikszentmihalyi 1977; Csikszentmihalyi and LeFevre 1989, p. 816, emphasis added) preceded by a set of antecedent conditions necessary for the experience to be achieved and followed by a set of consequences that occurs as a result of the process.

We define the *flow experience* in a CME as the state occurring during network navigation, which is (1) characterized by a seamless sequence of responses facilitated by machine interactivity, (2) intrinsically enjoyable, (3) accompanied by a loss of self-consciousness, and (4) self-reinforcing. In the flow experience, which formalizes and extends a sense of playfulness (Csikszentmihalyi 1977; Csikszentmihalyi and LeFevre 1989), consumers are so acutely involved in the act of network navigation in the hypermedia CME that "nothing else seems to matter" (Csikszentmihalyi 1990, p. 4).

Two primary antecedents must be present in sufficiently motivated users of a hypermedia CME for the flow experience to occur. Consumers must focus their attention on the interaction, narrowing their focus of awareness so that irrelevant perceptions and thoughts are filtered out, and they must perceive a balance between their skills and the challenges of the interaction.

The key consequences of the flow experience for consumers are increased learning, exploratory and participato-

ry behaviors, positive subjective experiences, and a perceived sense of control over their interactions in the hypermedia CME.

When in the flow state, irrelevant thoughts and perceptions are screened out and the consumer focuses entirely on the interaction. The flow experience involves a merging of actions and awareness, with concentration so intense that there is little attention left to consider anything else. A consumer's action in the flow state is experienced as a "unified flowing from one moment to the next, in which he is in control of his actions, and in which there is little distinction between self and environment, between stimulus and response, or between past, present and future" (Csikszentmihalyi 1977, p. 36). Self-consciousness disappears, the consumer's sense of time becomes distorted, and the resulting state of mind is extremely gratifying.

Flow has been previously noted as a useful construct for describing human-computer interactions (Csikszentmihalyi 1990; Ghani, Supnick, and Rooney 1991; Trevino and Webster 1992; Webster, Trevino, and Ryan 1993). Our conceptualization extends these ideas by explicitly developing the flow construct in the context of network navigation in a hypermedia CME and arguing that flow consists of a process state that requires a set of antecedents to occur and results in a set of consequences.

On first inspection, the flow experience may appear to be similar to attention and involvement, as well as other construct events (Privette and Bundrick 1987) of peak experience and peak performance. Privette's (1983) and Privette and Bundrick's (1987) distinction between flow, peak experience, and peak performance is helpful for understanding how flow differs from these related construct events, as well as from attention and involvement. Privette considers attention and involvement to be common qualities of flow, peak experience, and peak performance, but attention and involvement do not differentiate between the three. Flow is characterized by fun and occurs in structured activities in which action follows action. Peak performance is characterized by a clear focus, a strong sense of self (rather than loss of self, as in flow), and a sense of fulfillment, whereas peak experience is characterized by a transpersonal and spiritual quality that has much higher levels of experienced joy than flow has.

Flow is a central construct because of the nature of commercial activity in a CME such as the World Wide Web. It is important to understand that commercial activity on the Web consists of much more than purchasing products in on-line storefronts. Hoffman, Novak, and Chatterjee (1995) identify six functional categories of commercial activity on the Web: on-line storefronts, Internet presence sites, content sites, malls, incentive sites, and search agents. To date, Internet presence sites, rather than on-line storefronts, dominate commercial activity. Using the typology proposed by Hoffman, Novak, and Chatterjee, Kaul (1995) found that only 18% of a random sample of 290 Web sites served as on-line storefronts. The remaining 82% were informational or image-based Internet presence sites or directories of other commercial sites.

Internet presence sites either provide detailed information on a firm's offerings (e.g., the Web sites of Federal Express, Sun Microsystems, Volvo) or create an image and attempt to build an ongoing relationship with the consumer (e.g., the Web sites of Zima, Reebok, David Letterman). Internet presence sites are a new form of nonintrusive advertising, in which the customer actively chooses to visit and interact with the firm's marketing communication efforts. Recent efforts involve the merging of information and image in innovative ways (see, e.g., L. L. Bean's sophisticated "trail metaphor" Web site at www.llbean.com). Measures of duration time spent at an Internet presence site, depth of search through the site, navigation patterns through the site, and repeat visits to the site are crucial outcome measures for evaluating the effectiveness of such sites. Flow affects all of these outcome measures and is an important consideration for understanding consumer behavior in commercial Web sites.

The process model described subsequently has direct implications for understanding experiential behavior. The primary focus of this process model is on network navigation in integrated destination sites (Hoffman, Novak, and Chatterjee 1995), rather than on mechanisms for search and choice in such environments. Nevertheless, goal-directed processes of prepurchase search and choice in a CME take place against a background of network navigation and involve flow within and across Web sites; we address those issues, as well.

The Process Model

In Figure 6, we present a preliminary process model of network navigation in a hypermedia CME. For expository purposes, we diagram only the most important links. The model in Figure 6 shows neither the complex set of feedback loops and pathways nor the fully dynamic nature of the process. A consumer enters the hypermedia CME and engages in network navigation. There are several points of exit from the environment, as well as opportunities to continue navigation, with flow in essence serving as the glue holding the consumer in the hypermedia CME.

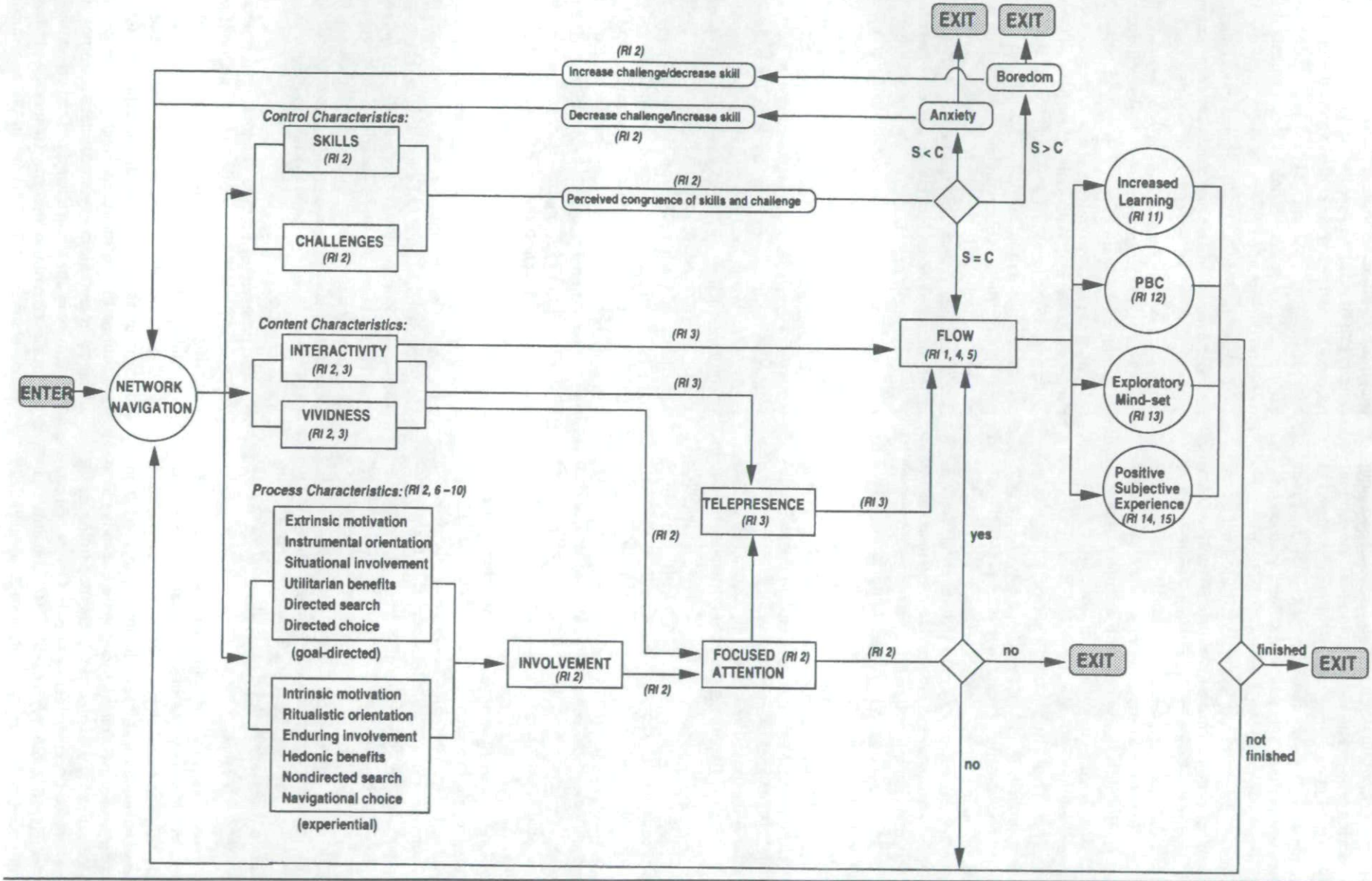
In the following section, we introduce a series of 15 research issues related to the components of the process model shown in Figure 6. We have mapped the research issues on the boxes and pathways of the Figure to provide an overview of how they relate to the process model. Of the first five research issues, the first three deal with flow measurement, whereas the next two deal with consumer heterogeneity and flow in CMEs. In the second five research issues, we discuss experiential and goal-directed behavior in a CME. In the final five research issues, we deal with the consequences of flow.

Research Issues

Flow Measurement

1. Measurement approaches. Reliable and valid measurement of flow is necessary for further development of the preliminary process model in Figure 6. We view flow in a CME as measurable along a continuum and note that flow can be inferred from its antecedents and consequences. Likert scales provide one approach to measuring flow. Ghani, Supnick, and Rooney (1991) and Ghani and Deshpande

FIGURE 6
A Model of Network Navigation in a Hypermedia CME



Note: RI = Research Issue; S = Skills; C = Challenges.

(1994) operationalized flow as a self-report scale containing items measuring enjoyment and concentration. Perceptions of control, skill, and challenge were significantly related to this flow measure. Trevino and Webster (1992) used four items to measure flow. The four items dealt with components of control, attention focus, curiosity, and intrinsic interest. Webster, Trevino, and Ryan (1993) used a 12-item scale expanded from Trevino and Webster's study to measure flow, again measuring the same four components of flow. Although these components correspond more to antecedents of flow than to the flow experience, the scales used were significantly correlated with a variety of consequences of flow, such as enjoyment, positive attitudes, and future usage intentions.

Privette (1983) and Privette and Bundrick (1987) take an alternative approach to measuring flow by asking respondents to describe construct events they have personally experienced involving flow, peak experience, peak performance, average events, misery, and failure. Brief descriptions of these construct events are provided, and respondents provide personal examples of and rate each construct event using Privette's (1984) experience questionnaire. Lutz and Guiry (1994) used a modified version of Privette's experience questionnaire in a study on consumption experiences.

The most widely used approach to measuring flow is the Experience Sampling Method (Csikszentmihalyi and Csikszentmihalyi 1988; Csikszentmihalyi and LeFevre 1989; Ellis, Voelkl, and Morris 1994; Mannell, Zuzanek, and Larson 1988). With the Experience Sampling Method, respondents are electronically paged at random intervals throughout the day, at which point they immediately complete a self-report experiential sampling form with open-ended items and rating scales. On a typical experiential sampling form, respondents rate the experience in which they are currently engaged on the challenge of the experience, their own skills in the experience, a series of items measuring affect and arousal, and items measuring motivation, concentration, and creativity.

A comprehensive flow measurement procedure for a CME must include measures of the antecedent conditions (see Research Issues 2 and 3), consequences (see Research Issues 11 through 15), and variables related to the psychological experience of flow (e.g., lose track of time, self-awareness, concentration, mood, control). We anticipate that qualitative research (such as verbal protocol statements obtained from consumers as they interact with a CME), process tracing measures of actual behavior during the interaction, and survey research (both traditional and electronic) will all be necessary to develop reliable and valid measures of flow in a CME.

2. *Primary antecedents of flow.* Two primary antecedent conditions are necessary for the flow state to be experienced: (1) skills and challenges must be perceived as congruent and above a critical threshold and (2) focused attention must be present. *Skills* are defined as the consumer's capacities for action, and *challenges* as the opportunities for action available to the consumer in a CME. Only when consumers perceive that the hypermedia CME contains challenges congruent with their own skills can flow potentially occur. If network navigation in a CME does not provide for congruence of skills and challenges, then consumers either become bored (i.e., their skills exceed the challenges) or anxious (i.e., the challenges exceed their skills) and either exit the CME or select a more (or less) challenging activity within it (see Figure 6).

Csikszentmihalyi's (1977) original model specified that flow occurred when an equal match between skill and challenge was perceived, regardless of whether skill and challenge were equally high or equally low. More recent reformulations (Csikszentmihalyi and Csikszentmihalyi 1988; Csikszentmihalyi and LeFevre 1989) specify that "flow results from experience contexts characterized by a match between challenge and skills only when both challenges and skills exceed the level that is typical for the day to day experiences of the individual" (Ellis, Voelkl, and Morris 1994, p. 338). Congruent skills and challenges that lie below the person's typical level are said to result in an apathy state (Csikszentmihalyi and Csikszentmihalyi 1988). Congruent, above-threshold skills and challenges as a prerequisite for flow are also consistent with the theory of an optimal stimulation level (e.g., Holbrook and Gardner 1993; Raju 1980; Steenkamp and Baumgartner 1992).

Considerable disagreement exists over how many channels, or categories, should be used to represent the various patterns of congruence or incongruence of high and low skills and challenges and how these channels should be labeled. Ellis, Voelkl, and Morris (1994) note that 3-channel (Csikszentmihalyi 1977), 4-channel (Csikszentmihalyi 1988), 8-channel (Csikszentmihalyi and Nakamura 1989; Massimini and Carli 1988), and 16-channel (Massimini and Carli 1988) models have been developed; recently, Clarke and Haworth (1994) proposed a 9-channel model. The labeling of these channels also can be inconsistent. For example, Csikszentmihalyi and Csikszentmihalyi (1988) define low skill and low challenges as *apathy,* whereas Clarke and Haworth (1994) call this combination *relaxation.*

There is also debate about the best way to determine whether skills or challenges exceed the critical threshold or typical level. Most often, within-subject standardization is used (e.g., Csikszentmihalyi and Csikszentmihalyi 1988; Csikszentmihalyi and LeFevre 1989). However, Ellis, Voelkl, and Morris (1994) note that within-subject standardization precludes studying individual differences, which have been found to be an important predictor of flow (see Research Issue 4). Consequently, some researchers (e.g., Clarke and Haworth 1994) do not employ within-subject standardization.

The presence of focused attention is also necessary to experience flow. *Focused attention* is characterized as "a centering of attention on a limited stimulus field" (Csikszentmihalyi 1977, p. 40). Webster, Trevino, and Ryan (1993) note that the computer screen functions as the limited stimulus field and that subjects report being "mesmerized" during their computer interactions. In Figure 6, we indicate the role of the content characteristics of vividness and interactivity in attracting attention. The performance characteristics of ease of use, mapping (naturalness of how human actions are connected to actions in the hypermedia CME;

Steuer 1992), speed, and range all combine to increase interactivity. *Vividness* is defined as "the representational richness of a mediated environment as defined by its formal features" (Steuer 1992, p. 81), such as its breadth and depth. *Breadth* refers to the number of sensory dimensions presented and is closely related to media concurrency (Valacich et al. 1993) and media richness (Daft and Lengel 1986; Daft, Lengel, and Trevino 1987). *Depth* is the resolution or the quality of the presentation (Steuer 1992) and is highly correlated with media bandwidth. In the hypermedia CME, both breadth and depth, in general, are high.

Extrinsic and intrinsic motivation (Csikszentmihalyi 1977, Chapter 1; Davis, Bagozzi, and Warshaw 1992; Graef, Csikszentmihalyi, and Gianinno 1983) are important process characteristics that affect focused attention through involvement (Zaichkowsky 1986). Davis, Bagozzi, and Warshaw (1992, p. 1112), in summarizing the motivational literature, note that extrinsic motivation applies to activities performed because they are instrumental to achieving a valued outcome, whereas intrinsic motivation applies to activities performed "for no apparent reinforcement other than the process of performing the activity." Extrinsic motivation creates situational self-relevance (Bloch and Richins 1983; Celsi and Olson 1988), whereas intrinsic motivation, or *autotelic activities* (Csikszentmihalyi 1977, Chapter 2), create intrinsic self-relevance. We define *involvement* as felt involvement (Celsi and Olson 1988), which is formed by the presence of situational and/or intrinsic self-relevance. Felt involvement affects attention and comprehension effort (Celsi and Olson 1988). In the context of human-computer interaction, Webster, Trevino, and Ryan (1993) find significant positive correlations between factors for intrinsic interest and/or curiosity and focused attention.

3. Secondary antecedents of flow. Because of the presence of focused attention and perceived congruence of skills and challenges, two additional antecedents—interactivity and telepresence—enhance flow. These additional antecedents increase the subjective intensity of the consumer's flow state, though they are not sufficient alone to produce a flow state. A strong sense of telepresence (i.e., the mediated perception of an environment) is induced by vividness and interactivity (Sheridan 1992), as well as focused attention. Although the user of a hypermedia CME always experiences some level of telepresence and interactivity, higher levels can boost flow. Feedback loops, however, also must be considered. When telepresence is too high—for example, when the medium is too "hot" (McLuhan, 1964, p. 36)—challenges may become greater than skills and flow cannot be achieved (Steuer 1992).

For marketers, the primary and secondary antecedents of flow suggest that flow is not a constant state. Consumers move in and out of flow—as a function of control, content, and process characteristics. To some extent, all these characteristics may be influenced by marketing activities. The congruence of skills and challenges is something that can be facilitated by interface design. For example, a user-specified difficulty level can be designed to avoid anxiety in novice users and boredom in experienced users. Because lack of congruence may lead the consumer to exit the CME, it is important to provide opportunities for consumers to actively select activities that create congruence. Content characteristics such as interactivity and vividness lead to telepresence and can be affected directly through product design considerations. Process characteristics, specifically the distinction between extrinsically and intrinsically motivated consumers, are an important segmentation basis.

Consumer Heterogeneity and Flow in CMEs

4. The autotelic personality. Consumers are heterogeneous in their ability to experience flow in a CME. Csikszentmihalyi (1977, pp. 21–22) distinguishes between autotelic experiences, activities, and personalities. Flow is the autotelic experience; autotelic activities are those activities relatively more likely to lead to flow; and the autotelic personality trait characterizes a person "who is able to enjoy what he is doing regardless of whether he will get external rewards from it" (Csikszentmihalyi 1977, p. 22) and who thus is more likely to experience flow for a given activity. Csikszentmihalyi (1988) later suggests that the ability to experience flow may be learned in the family, developed through practice, or even have a neurological component. Whatever the underlying explanation for the individual differences, Ellis, Voelkl, and Morris (1994) find that considerable variation in subjective experience was explained by variables that indicated the autotelic nature of the person, over and above the variance explained by the perceived congruence of skills and challenges. For marketers, the crucial implication is that the relative likelihood a person will experience flow is an important segmentation basis in a CME.

5. Optimal stimulation level and flow. People with a higher optimal stimulation level (OSL) are more likely to possess the autotelic personality trait. The fundamental idea of OSL theories (e.g., Zuckerman 1979) is that an intermediate level of stimulation obtained from the environment corresponds to the most favorable affective reaction. Furthermore, there are individual differences in the OSL—with people having a higher OSL exhibiting increased curiosity-motivated, variety-seeking, risk-taking (Steenkamp and Baumgartner 1992), and exploratory behavior (Raju 1980). Holbrook and Gardner (1993) found the duration time of consumption of an experiential good to be related to an optimal arousal level, which has clear implications for experiential consumer behavior in a CME.

As we previously discussed, flow in a CME requires the congruence of above-threshold skills and challenges. Yet, people with low optimal stimulation levels are more likely to experience anxiety in their initial interactions with a CME, as they will be overwhelmed by the wide variety of options available to them and will perceive their skills to be well below the uncomfortably high stimulation level of the CME (i.e., the challenges of the task). Furthermore, people with low OSLs who have mastered the basics of network navigation also are more likely to later become bored with a CME, as they will be relatively less likely to seek out sufficiently high challenges to maintain congruence with their increasing skills. Thus, we anticipate that people with higher OSLs are more likely to possess the autotelic personality trait and experience flow in a CME.

Experiential and Goal-Directed Behavior in a CME

6. Definition of experiential and goal-directed behavior. Two broad categories of behavior in which consumers engage during time spent in a CME are goal-directed and experiential behavior. Goal-directed and experiential behavior are characterized respectively by (1) extrinsic versus intrinsic motivation, (2) instrumental versus ritualized orientation, (3) situational versus enduring involvement, (4) utilitarian versus hedonic benefits, (5) directed versus nondirected search, and (6) goal-directed versus navigational choice.

For example, the corporate buyer using the Web to close a deal for computer components experiences an extrinsically motivated, instrumental, goal-directed flow state. On the other hand, net surfers exploring the Web in their daily quest for the latest interesting sites experience an intrinsically motivated, ritualized, experiential flow state. It is important to recognize that consumers engage in both goal-directed and experiential behaviors, flow may occur with both types of behaviors, and the optimal design of a CME site differs according to whether the behavior is goal-directed or experiential.

In the communications literature, a distinction is drawn between instrumental and ritualized orientations to media (Rubin 1984; Rubin and Perse 1987). Ritualized orientations focus "more on the medium, rather than on particular content," are "associated with diffuse motives (e.g., pass time, habit, relaxation)," and are a "less intentional and nonselective orientation, a time-filling activity" (Rubin and Perse 1987, p. 59). In contrast, instrumental orientations are "more intentional and selective," which reflects "purposive exposure to specific content" (p. 59). The distinction between instrumental and ritualized orientations bears considerable resemblance to the classification of expected benefits in the marketing literature into utilitarian and hedonic or experiential benefits (e.g., Havlena and Holbrook 1986; Srinivasan 1987), which as we previously noted are, respectively, extrinsically and intrinsically motivated.

We also can differentiate flow states resulting from goal-directed and experiential behaviors on the basis of involvement and search behavior. Most commercial Web sites currently provide information rather than offer an opportunity to purchase a product. Thus, much of today's consumer search activities are more likely to involve what has been termed *ongoing search* than *prepurchase search* (Bloch, Sherrell, and Ridgway 1986, p. 121). Similarly, Biehal and Chakravarti (1982, 1983) distinguish between directed and nondirected learning. Bloch, Sherrell, and Ridgway (1986) note that ongoing search is a function of enduring involvement with the product (or with the CME), whereas prepurchase search is a function of situational involvement with the purchase.

For choice, in experiential behavior, "activities are not guided by goals or outcomes, but by the process itself" (Bloch, Sherrell, and Ridgway 1986, p. 121). Similarly, Deci and Ryan (1985, p. 155) note that for intrinsically motivated people engaged in flow, choice is "intuitive and spontaneous" and does not involve conscious, deliberate decisions. Consumer choice in goal-directed behavior is based on a clearly definable goal hierarchy, and movement through this goal hierarchy involves choices among products and services, information sources, and navigational alternatives. Consumer choice in experiential behavior is dominated by choices among navigational alternatives and corresponds to a relatively unstructured and continually changing goal hierarchy.

7. Search motives and involvement for experiential and goal-directed behavior. Goal-directed versus experiential behaviors depend on distinct search motives combined with the object of involvement:

Search Motive	Involvement	Behavior
Task completion	Situational involvement with **goal** →	Goal-Directed
Prepurchase deliberation	Situational involvement with **product** →	Goal-Directed
Build information bank	Enduring involvement with **product** →	Experiential
Opinion leadership	Enduring involvement with **product** →	Experiential
Recreation	Enduring involvement with **process** →	Experiential

Goal-directed behaviors are characterized by situational involvement and directed search. In general, a CME user is involved with a specific task-completion goal. Marketers are particularly concerned with understanding the prepurchase deliberations for product purchase goals. Search motives in experiential behaviors, however, are more varied. When the consumer exhibits enduring involvement with a product or product category, Bloch, Sherrell, and Ridgway (1986), Biehal and Chakravarti (1982, 1983), and Bettman (1979) indicate that consumers may search to build an information bank or knowledge base in their memories for potential future use. In addition, Richins and Root-Shaffer (1988) find that enduring involvement (occurring in experiential flow) led to opinion leadership, whereas situational involvement did not. Thus, an opinion leader may be motivated to search and engage in experiential behaviors to disseminate product news, advice, and personal experience by word of mouth. Moreover, consider that consumers may be involved with the CME itself. Bloch (1995) suggests that there likely are two segments of consumers who navigate a CME: (1) those who exhibit enduring involvement with an interest area and (2) those who are navigating because they exhibit enduring involvement with computers. We propose that the latter segment is engaged in nondirected search for recreational purposes (Bloch, Sherrell, and Ridgway 1986; Csikszentmihalyi 1983).

Thus, experiential behavior is relevant for (1) word-of-mouth strategies based on influencing opinion leaders, (2) providing entertainment and recreation, and (3) enhancing consumers' product knowledge, whereas goal-directed behavior is relevant for task-specific use of a CME, such as prepurchase deliberation. This distinction is critical and suggests that marketers should take care to focus not only on goal-directed behaviors in a CME (e.g., product purchase),

but also on nondirected experiential behaviors (e.g., net surfing), which are strategically important as well.

8. External memory. A CME can provide devices for external memory: User-generated "bookmark files" containing custom lists of consumers' favorite Web sites are cases in point, as is comparative information about brands (e.g., product reviews), which is directly retrievable through an on-line storefront. Although information gained from goal-directed search activity is more accessible in memory than that from experiential search activity, CMEs provide devices for external memory that negate this advantage. Biehal and Chakravarti (1983) find that information retrieved from directed search (goal-directed) had greater memory accessibility than information retrieved from nondirected search (experiential). Furthermore, the likelihood of choosing a brand for which information was acquired during directed search was not significantly different than the likelihood of choosing a brand for which information was externally available; in both situations, the choice likelihood was greater than when brand information was acquired through nondirected search. However, the external memory devices present in CMEs imply that information retrieved from nondirected search should have a greater impact on subsequent consumer choice behavior than that in traditional environments, because the external memory devices are used by the consumer.

9. Choice behavior and decision making in a CME. We also distinguish goal-directed and experiential behavior on the basis of consumer decision and choice processes. Most research on consumer choice and decision making in marketing deals with activities directly involving or motivated by product purchase (e.g., Payne, Bettman, and Johnson 1993). Although theories of problem-solving behavior have been developed for extensive, limited, and routine problem-solving scenarios, all three scenarios are typically motivated by a purchase outcome. In a CME, however, the process of network navigation continually confronts the consumer with an ongoing series of decisions that are potentially but not necessarily related to a purchase outcome or other task completion. Although existing models of adaptive decision making can explain much of goal-directed choice in a CME, they hold relatively lesser applicability to navigational choice in flow occurring from experiential behavior. In addition, CMEs provide devices that augment the individual decision processes with input from communal or machine agents.

Choice in experiential behavior primarily involves an ongoing series of navigational decisions of what to do next. This is a highly unstructured activity and, as was noted previously, corresponds to an intrinsically motivated, ritualized, hedonic use of the CME. Nonetheless, a key assumption of the nature of adaptive decision behavior expected in goal-directed situations—"that people are motivated to use as little effort as necessary to solve a problem" (Payne, Bettman and Johnson 1993, p. 13)—would realistically hold in nondirected experiential situations as well. Thus, in deciding among navigational alternatives, heuristics and noncompensatory decision rules most likely would be applied in experiential activities. Relatively little research, however, deals with choice behavior during experiential activities, so we must be cautious and not overly dependent on goal-directed decision strategies to provide explanations of consumer choice. We liken flow from experiential behaviors to navigating through an amusement park where the visitor is presented with a wide array of ride choices in a nonlinear fashion.

For flow occurring with experiential behaviors, outcome variables other than choice should be investigated. For example, Holbrook and Gardner (1993) argue that duration time is a critical outcome measure of consumption experiences, and Olney, Holbrook, and Batra (1991) use viewing time as a dependent variable in a model of advertising effects. Indeed, duration times are a useful behavioral indicator of experiential versus goal-directed orientations. For example, 43% of all calls to the French Minitel system are for the electronic telephone directory (a goal-directed activity) but account for only 21% of the total connect time. On the other hand, only 6% of all calls to Minitel are for the chat services (an experiential activity), but they account for 15% of the total connect time (Cats-Baril and Jelassi 1994).

In structured decision environments, such as an on-line storefront offering a variety of goods, the strategies typically applied by consumers in traditional environments (e.g., Payne, Bettman, and Johnson 1993, Chapter 2) are likely to be used. These strategies, however, will be augmented by powerful decision aids (e.g., Payne, Bettman, and Johnson 1993, Chapter 7) that are feasible only in a CME. At the most basic level, these decision aids involve information displays that increase the consumer's processing capacity, whereas more sophisticated approaches use decision support systems to assist the decision maker. Both of these categories of decision aids are local, rather than network-based, because they involve only information presented at a given location in a CME.

Intelligent interface agents (e.g., Maes 1994) provide a network-based decision aid, which is particularly useful for vastly enhancing the consumers' ability to perform search activities by reducing information overload. An example of an agent present on the World Wide Web is Digital Research Laboratory's Alta Vista search engine (www.altavista.digital.com), a robot program that automatically traverses the World Wide Web in search of hypermedia documents and stores and categorizes them in an extensive and rapidly growing database that can be accessed by an intelligent search interface. As of March 1996, the Alta Vista database can access 11 billion words found in over 22 million Web pages.

Although agents have been portrayed in the popular media as engaging in semiautonomous, high-level, human-like interactions, "the most successful interface agents are those that do not prohibit the user from taking actions and fulfilling tasks personally" (Maes 1994, p. 31). Thus, agents can serve as effective decision aids in an information intensive environment. One promising approach in a networked environment such as a CME are collaborative interface agents (Lashkari, Metral, and Maes 1994), which combine decision aids across a segment of similar consumers, rather than rely on a single consumer's input.

10. Developmental patterns of flow states. We expect that experiential behavior dominates a user's early flow ex-

periences in a CME, but that over time goal-directed behavior also leads to flow experiences. Early interactions in the hypermedia CME that lead to flow are characterized by a nondirected, time-passing, ritualized quality. Over time, ritualized use evolves into instrumental use as consumers accumulate experience navigating within the medium. A greater degree of technical skill is required to successfully perform goal-directed behaviors than experiential behaviors. Experiential behaviors, in which the consumer experiments and becomes familiar with the CME, are accompanied by an increase in skills that the consumer develops to meet the challenges presented by the environment. In other words, learning occurs and consumers begin to seek greater challenges. Thus, an "instrumental" orientation is likely to dominate a consumer's later interactions in the environment, though both orientations may be present at different points in time, depending on consumer characteristics.

This discussion has important implications for the adoption of CMEs, particularly for those people seeking the "killer application." Developmental patterns of flow states suggest that goal-directed behaviors such as home shopping and home banking do not necessarily lead to flow for new users of a CME. On the other hand, experiential behaviors, such as browsing on-line magazines, participating in interactive chat rooms, and exploring a Web corporate home page for a topic with which the consumer exhibits enduring involvement, would be more likely to lead to flow and thus stimulate adoption in new users.

The Consequences of Flow

In the final set of research issues, we address consequences of flow. Based on the existing literature, consumers who experience flow in a hypermedia CME achieve increased learning, increased perceived behavioral control (PBC), increased exploratory and participatory behavior, and positive subjective experiences. In addition, we briefly consider potentially negative consequences of flow. Research Issues 11 through 15, seen in the context of the network navigation model in Figure 6, contrast consumers who experience the flow state versus those who do not. These research issues apply to both heterogeneity *across* consumers according to their ability to experience flow (Research Issue 4) and heterogeneity *within* consumers according to whether flow is experienced (Research Issues 2 and 3). Thus, flow is viewed as both a trait and a state, much as Webster and Martocchio (1992) view playfulness as both a trait and a state. Note that other researchers have viewed positive mood as both a trait and a state (e.g., George 1991).

11. Consumer learning. Consumers who experience the flow state are more likely to retain more of what they perceive than consumers who do not. Playfulness (Webster and Martocchio 1992) and flow (Webster, Trevino, and Ryan 1993) have been found to relate to learning. Early ritualized use of a CME (i.e., scanning, exploring, and wandering; see Canter, Rivers, and Storrs 1985) facilitates the learning needed to progress to instrumental usage involving browsing and searching (i.e., purposive search behavior to identify on-line vendors selling products or services desired by the consumer). Webster, Trevino, and Ryan (1993) suggest that because consumers develop and apply their abilities through exploratory behaviors that characterize flow interactions, learning is a reasonable outcome of the flow state.

The distinction between directed and nondirected learning (Bettman 1979; Biehal and Chakravarti 1982, 1983) further differentiates flow occurring in goal-directed and experiential behaviors. In experiential behavior, ongoing, nondirected search produces latent learning (Hilgard and Bower 1966, pp. 199–200), in which the consumer learns about the environment, "even if the specific knowledge gained has no direct relevance to current purchases" (Bettman 1979, p. 88). In goal-directed behavior, a product choice decision or specific task completion is the primary goal for learning. Consumer learning suggests that flow resulting from goal-directed behaviors leads to more informed decisions, whereas flow resulting from experiential behaviors leads to greater recall and word-of-mouth activities.

12. Perceived behavioral control. Consumers who experience the flow state in a hypermedia CME are expected to have greater PBC than those who do not. *Perceived behavioral control* is defined as "the perceived ease or difficulty of performing the behavior and ... is assumed to reflect past experience as well as anticipated impediments and obstacles" (Azjen 1988, p. 132). Perceived behavioral control is a component of Azjen's (1988, 1991) theory of planned behavior, extends the well-known theory of reasoned action (Azjen and Fishbein 1980), and is useful for predicting behaviors that are not under complete volitional control. Similar to the original expectancy-value model, the theory of planned behavior posits that conceptually independent determinants of attitudes toward the behavior, subjective norms with respect to the behavior, and a new component—*perceived control over the behavior*—affect behavioral intentions. In addition, PBC directly affects behavior. Thus, usage of a hypermedia CME (for those people with the prerequisite technology) depends jointly on motivation (intentions) and ability (behavioral control).

Perceived behavioral control (cf. Bandura's [1977, 1982] earlier notion of *perceived self-efficacy*) also can be interpreted as a confidence construct and is an important determinant of CME usage, because behavior in a CME is strongly influenced by consumers' confidence in their ability to engage in network navigation. A consumer's perception of behavioral control over CME use and its impact on intentions and actions is more important than real control (Azjen 1988).

Flow significantly correlates with perceived control (Ghani, Supnick, and Rooney 1991). Webster, Trevino, and Ryan (1993) define control as one of four dimensions of flow and find that it significantly correlated with two of the other three dimensions (curiosity and intrinsic interest, but not attention focus). Webster and Martocchio (1992) find a significant correlation between microcomputer playfulness and self-rated computer competence. Because the hypermedia CME is, first and foremost, an interactive environment, it affords the foundation for consumer control that is impossible in traditional, passive media. Control comes from both consumers' perception of their ability to adjust the CME and their perception of how the CME responds to

their input, with consumer adjustment taking the form of network navigation.

We here imply that flow resulting from experiential behavior results in increased PBC over network navigation, whereas flow resulting from goal-directed behavior results in increased PBC over task completion. Because the former is a prerequisite for the latter, experiential behavior flow facilitates the learning necessary to "move up" to goal-directed activities.

13. Exploratory behavior. We expect that consumers who experience the flow state in a hypermedia CME exhibit more exploratory behaviors than those who do not. Webster, Trevino, and Ryan (1993) found that flow correlated positively with experimentation and perceptions of software flexibility and modifiability. Other researchers have found that higher levels of playfulness or flow in human-computer interactions correlate with higher experimentation (Ghani and Deshpande 1994; Ghani, Supnick, and Rooney 1991; Katz 1987). This argues for a flexible hypermedia environment that encourages exploratory behavior on the part of consumers. Implications of increased exploratory behaviors are increased risk taking in goal-directed behaviors (Raju 1980; Steenkamp and Baumgartner 1992) and broader exposure to content in experiential behaviors. Furthermore, increased exploratory use has been found to lead to greater extent of use (Ghani and Deshpande 1994).

14. Positive subjective experiences. We expect that consumers who experience the flow state in a hypermedia CME exhibit more positive subjective experiences than those who do not. Webster, Trevino, and Ryan (1993, p. 412) note that "higher playfulness results in immediate subjective experiences such as positive mood and satisfaction" (Csikszentmihalyi 1977; Levy 1983; McGrath and Kelly 1986). Previous research on human-computer interactions (Sandelands and Buckner 1989; Starbuck and Webster 1991; Webster and Martocchio 1992) has shown that higher degrees of pleasure and involvement during computer interactions lead to concurrent subjective perceptions of positive affect and mood. A study conducted by Gardner, Dukes, and Discenza (1993) shows that the more people use computers, the more their self-confidence with respect to computers increases. This in turn causes more favorable attitudes toward computers. Csikszentmihalyi (1977 p. 36) notes that "people seek flow primarily for itself;" thus, flow itself serves as a positive reinforcer, which increases the probability of further use of a hypermedia CME.

Together, positive subjective experiences and the increased PBC resulting from flow feed back into the planned behavior model discussed previously and increase the probability that the CME will be used in the future. Webster, Trevino, and Ryan (1993) find flow to be positively correlated with expectations of future voluntary computer interactions. Thus, we anticipate that the positive affect generated by flow translates into longer duration time spent visiting a CME and increased repeat visits.

15. Distortion in time perception. The positive subjective experience of flow also has been linked to a distortion in time perception (Csikszentmihalyi 1977), whereby the consumer is unaware of the passage of time. Hauser, Urban, and Weinberg (1993) provide a model of how, in the presence of time pressure, the allocation of search time to negative information sources increases, whereas in the absence of time pressure, the allocation of search time to positive information sources increases. Thus, to the extent that the flow state reduces time pressure and increases time spent at the site, it contributes to a relatively greater amount of time being allocated to searching for positive information sources. Because of the tendency of computer-mediated environments to encourage negative word-of-mouth communications, actions that marketers can take to facilitate positive search behavior become particularly important.

Negative consequences of flow. There are some potentially negative consequences of flow that must be considered. Because flow can be its own reward, consumers may explore a CME for its own sake, rather than purposely search for specific information. Thus, too much flow may distract the consumer from purchase-related activities. As Webster, Trevino, and Ryan (1993, p. 422) note, playfulness may produce longer time to task completion; at an extreme, "playful computer systems may be so enjoyable that employees neglect other tasks." However, if one marketing objective of a hypermedia CME is to encourage the consumer to spend time at the site examining product-related information, for example, then this is not necessarily a problem. Flow has also been linked to overinvolvement (Csikszentmihalyi 1977), which leads to mental and physical fatigue. A related source of cognitive fatigue stems from the overwhelming complexity inherent in global hypermedia content (Gygi 1990).

Discussion and Conclusion

The hypermedia CME represents a fundamentally different environment for marketing activities than do traditional media and so-called interactive multimedia. The many-to-many communication model turns traditional principles of mass media advertising (based on the one-to-many model of Figure 1) inside out, rendering impossible the blind application of marketing and advertising approaches that assume a passive, captive consumer. Thus, marketers must carefully consider the ways in which advertising and communication models can be adapted and reconstructed (e.g., see Hoffman and Novak 1996) for the interactive, many-to-many medium depicted in Figure 3. In that new communication model, consumers can actively choose whether to approach firms through their Web sites and exercise unprecedented control over the management of the content with which they interact.

The opportunity for customer interaction in the hypermedia CME also is unprecedented. This can be utilized in numerous ways, including (1) the design of new products, (2) the development of product and marketing strategy, and (3) the innovation of content. The evolution of content in a hypermedia CME depends on not only the evolution of existing metaphors and communication codes from traditional media, but also new techniques and conventions inherent in the possibilities of the medium itself (Biocca 1992). One implication of this is that the content that makes hypermedia

CMEs commercially successful likely has not been invented yet and may require more than a simple continuous innovation of existing content (Grossman 1994).

Because consumers vary in their ability to achieve flow, new bases for market segmentation are needed for marketing in hypermedia CMEs. Scholars must determine the variables that relate to a consumer's propensity to enter the flow state. Such information can be used to develop marketing efforts designed to maximize the chances of the consumer entering the flow state. Because we believe that repeat consumption behavior, that is, repeat visits to a hypermedia CME, are increased if the environment facilitates the flow state, the marketing objective at trial must provide for these flow opportunities.

In summary, the new medium-as-market represented by the hypermedia CME—of which the World Wide Web on the Internet currently stands as the preeminent prototype—offers a working example of a many-to-many communication model in which the consumer is an active participant in an interactive exercise of multiple feedback loops and highly immediate communication. As such, it offers dynamic potential for growth and development, as well as a virtual revolution in both the way marketing academics and practitioners alike approach the problem of effective, consumer-oriented marketing in emerging media environments.

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